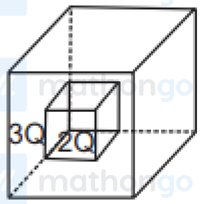
Q1 - 2024 (01 Feb Shift 1)

Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle θ with each other. When suspended in water the angle remains the same. If density of the material of the sphere is $1.5\ \mathrm{g/cc}$, the dielectric constant of water will be_____

(Take density of water $= 1\ \mathrm{g/cc}$)

Q2 - 2024 (01 Feb Shift 2)

C_1 and C_2 are two hollow concentric cubes enclosing charges 2Q and 3Q respectively as shown in figure. The ratio of electric flux passing through C_1 and C_2 is:



(1) 2 : 5

(2) 5 : 2

(3) 2 : 3

(4) 3 : 2

Q3 - 2024 (01 Feb Shift 2)

Suppose a uniformly charged wall provides a uniform electric field of $2 \times 10^4\ \mathrm{N/C}$ normally. A charged particle of mass 2 g being suspended through a silk thread of length 20 cm and remain stayed at a distance of 10 cm from the wall. Then the charge on the particle will be $\frac{1}{\sqrt{x}}\mu\mathrm{C}$ where $\mathrm{x} =$ _____. [use $\mathrm{g} = 10\ \mathrm{m/s^2}$]

Q4 - 2024 (27 Jan Shift 1)

An electric charge $10^{-6}\mu\mathrm{C}$ is placed at origin $(0,0)$ m of $\mathrm{X}-\mathrm{Y}$ co-ordinate system. Two points P and Q are situated at $(\sqrt{3},\sqrt{3})\mathrm{m}$ and $(\sqrt{6},0)\mathrm{m}$ respectively. The potential difference between the points P and Q will be :

(1) $\sqrt{3}$ V

(2) $\sqrt{6}$ V

(3) 0 V

(4) 3 V

Q5 - 2024 (27 Jan Shift 1)

A thin metallic wire having cross sectional area of 10^{-4} m^2 is used to make a ring of radius 30 cm. A positive charge of 2πC is uniformly distributed over the ring, while another positive charge of 30 pC is kept at the centre of the ring. The tension in the ring is ______ N; provided that the ring does not get deformed (neglect the influence of gravity). (given, $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$SI units)

Q6 - 2024 (27 Jan Shift 2)

Given below are two statements : one is labelled a

Assertion (A) and the other is labelled as

Reason(R)

Assertion (A) : Work done by electric field on moving a positive charge on an equipotential surface is always zero.

Reason (R) : Electric lines of forces are always perpendicular to equipotential surfaces.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) ((A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

Q7 - 2024 (27 Jan Shift 2)

Two charges of -4μC and $+4\mu$C are placed at the points $A(1,0,4)$m and $B(2,-1,5)$m located in an electric field $\overrightarrow{\mathrm{E}} = 0.20\hat{\imath}\,\mathrm{V/cm}$. The magnitude of the torque acting on the dipole is $8\sqrt{\alpha} \times 10^{-5}$Nm, Where $\alpha =$ _____

Q8 - 2024 (27 Jan Shift 2)

The electric potential at the surface of an atomic nucleus $(\mathrm{z}=50)$ of radius 9×10^{-13} cm is _____ $\times 10^{6}$ V

Q9 - 2024 (29 Jan Shift 1)

Two charges of $5\mathrm{Q}$ and $-2\mathrm{Q}$ are situated at the points $(3a, 0)$ and $(-5a, 0)$ respectively. The electric flux through a sphere of radius ' 4a ' having center at origin is :

(1) $\frac{2Q}{\varepsilon_0}$
(2) $\frac{5Q}{\varepsilon_0}$
(3) $\frac{7Q}{\varepsilon_0}$
(4) $\frac{3Q}{\varepsilon_0}$

Q10 - 2024 (29 Jan Shift 1)

Match List I with List II

	List I		List II
A.	$\oint \vec{B} \cdot \vec{dl} = \mu_0 i_c + \mu_0 \varepsilon_0 \frac{d\phi_E}{dt}$	I.	Gauss' law for electricity
B.	$\oint \vec{E} \cdot di = \frac{d\phi_B}{dt}$	II.	Gauss' law for magnetism
C.	$\oint \vec{E} \cdot \overrightarrow{dA} = \frac{Q}{\varepsilon_0}$	III.	Faraday law
D.	$\oint \vec{B} \cdot \overrightarrow{dA} = 0$	IV.	Ampere - Maxwell law

Chose the correct answer from the options given below

(1) A-IV, B-I, C-III, D-II

(2) A-II, B-III, C-I, D-IV

(3) A-IV, B-III, C-I, D-II

(4) A-I, B-II, C-III, D-IV

Q11 - 2024 (29 Jan Shift 1)

An electron is moving under the influence of the electric field of a uniformly charged infinite plane sheet S having surface charge density $+\sigma$. The electron at $t=0$ is at a distance of $1 \mathrm{~m}$ from S and has a speed of $1 \mathrm{~m/s}$. The maximum value of σ if the electron strikes S at $\mathrm{t}=1 \mathrm{~s}$ is $\alpha\left[\frac{\mathrm{m} \epsilon_0}{\mathrm{e}}\right] \frac{\mathrm{C}}{\mathrm{m}^2}$ the value of α is

Q12 - 2024 (29 Jan Shift 2)

An electric field is given by $(6 \hat{i}+5 \hat{j}+3 \hat{k}) N / C$. The electric flux through a surface area $30 \hat{\mathrm{i}} \mathrm{m}^2$ lying in YZ-plane (in SI unit) is :

(1) 90

(2) 150

(3) 180

(4) 60

Q13 - 2024 (30 Jan Shift 1)

The electrostatic potential due to an electric dipole at a distance ' r ' varies as:

(1) r

(2) $\frac{1}{r^2}$

(3) $\frac{1}{r^3}$

(4) $\frac{1}{r}$

Q14 - 2024 (30 Jan Shift 2)

A particle of charge ' $-q$ ' and mass ' m ' moves in a circle of radius ' r ' around an infinitely long line charge of linear density ' $+\lambda$ '. Then time period will be given as:

(Consider k as Coulomb's constant)

(1) $T^2=\frac{4\pi^2 m}{2k\lambda q}r^3$

(2) $T=2\pi r\sqrt{\frac{m}{2k\lambda q}}$

(3) $T=\frac{1}{2\pi r}\sqrt{\frac{m}{2k\lambda q}}$

(4) $T=\frac{1}{2\pi}\sqrt{\frac{2k\lambda q}{m}}$

Q15 - 2024 (30 Jan Shift 2)

Two identical charged spheres are suspended by string of equal lengths. The string make an angle of 37° with each other. When suspended in a liquid of density $0.7\ g/cm^3$, the angle remains same. If density of material of the sphere is $1.4\ g/cm^3$, the dielectric constant of the liquid is_____ $\left(\tan 37^\circ=\frac{3}{4}\right)$.

Q16 - 2024 (31 Jan Shift 1)

Two charges q and $3q$ are separated by a distance ' r ' in air. At a distance x from charge q, the resultant electric field is zero. The value of x is :

(1) $\frac{(1+\sqrt{3})}{r}$

(2) $\frac{r}{3(1+\sqrt{3})}$

(3) $\frac{r}{(1+\sqrt{3})}$

(4) $r(1+\sqrt{3})$

Q17 - 2024 (31 Jan Shift 2)

Force between two point charges q_1 and q_2 placed in vacuum at ' r ' cm apart is F. Force between them when placed in a medium having dielectric $K=5$ at ' $r/5$ ' cm apart will be:

(1) $F/25$

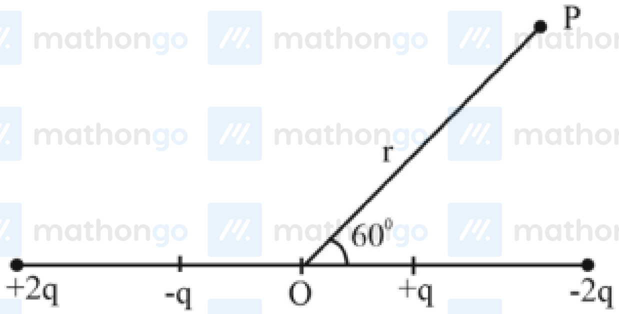
(2) $5\ F$

(3) $F/5$

(4) $25\ F$

Q18 - 2024 (31 Jan Shift 2)

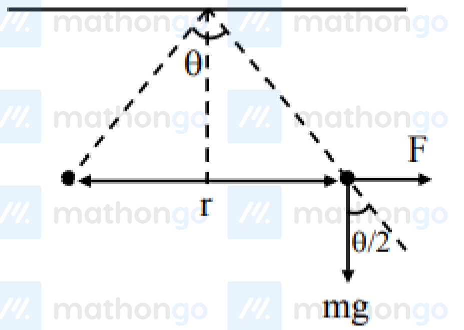
The distance between charges $+\mathrm{q}$ and $-\mathrm{q}$ is $2l$ and between $+2\mathrm{q}$ and $-2\mathrm{q}$ is $4l$. The electrostatic potential at point P at a distance r from centre O is $-\alpha\left[\frac{ql}{r^2}\right] \times 10^9$ V, where the value of α is_____ . (Use $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \mathrm{Nm}^2\mathrm{C}^{-2}$)



Answer Key

Q1 (3)	**Q2** (1)	**Q3** (3)	**Q4** (3)
Q5 (3)	**Q6** (4)	**Q7** (2)	**Q8** (8)
Q9 (2)	**Q10** (3)	**Q11** (8)	**Q12** (3)
Q13 (2)	**Q14** (2)	**Q15** (2)	**Q16** (3)
Q17 (2)	**Q18** (27)		

Q1



In air $\tan\frac{\theta}{2}=\frac{F}{mg}=\frac{q^2}{4\pi\varepsilon_0 r^2 mg}$

In water $\tan\frac{\theta}{2}=\frac{F'}{mg'}=\frac{q^2}{4\pi\varepsilon_0\varepsilon_r r^2 mg_{eff}}$

Equate both equations

$$\varepsilon_0\ g=\varepsilon_0\varepsilon_r g\left[1-\frac{1}{1.5}\right]$$

$$\varepsilon_r=3$$

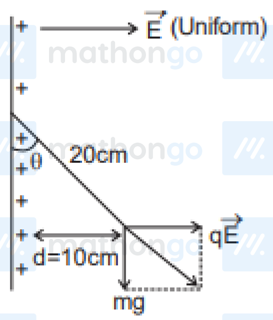
Q2

$$\phi_{\text{smaller cube}}=\frac{2Q}{\epsilon_0}$$

$$\phi_{\text{bigger cube}}=\frac{5Q}{\epsilon_0}$$

$$\frac{\phi_{\text{smaller cube}}}{\phi_{\text{bigger cube}}}=\frac{2}{5}$$

Q3



$\sin\theta = \frac{10}{20} = \frac{1}{2}$

$\theta = 30^\circ$

$\tan\theta = \frac{qE}{mg}$

$\tan 30^\circ = \frac{q \times 2 \times 10^4}{1 \times 10^{-3} \times 10}$

$\frac{1}{\sqrt{3}} = q \times 10^6$

$q = \frac{1}{\sqrt{3}} \times 10^{-6} C$

$x = 3$

Q4

Potential difference $= \frac{KQ}{r_1} - \frac{KQ}{r_2}$

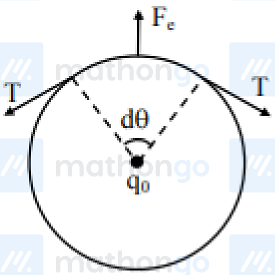
$r_1 = \sqrt{(\sqrt{3})^2 + (\sqrt{3})^2}$

$r_2 = \sqrt{(\sqrt{6})^2 + 0}$

As $r_1 = r_2 = \sqrt{6}$ m

So potential difference $= 0$

Q5



$$2\,\mathrm{T}\sin\frac{d\theta}{2}=\frac{\mathrm{kq}_0}{\mathrm{R}^2}\cdot\lambda\mathrm{R}d\theta$$

$$\left[\lambda=\frac{\mathrm{Q}}{2\pi\mathrm{R}}\right]$$

$$\Rightarrow \mathrm{T}=\frac{\mathrm{Kq}_0\mathrm{Q}}{\left(\mathrm{R}^2\right)\times 2\pi}$$

$$=\frac{\left(9\times 10^9\right)\left(2\pi\times 30\times 10^{-12}\right)}{(0.30)^2\times 2\pi}$$

$$=\frac{9\times 10^{-3}\times 30}{9\times 10^{-2}}=3\ \mathrm{N}$$

Q6

Electric line of force are always perpendicular to equipotential surface so angle between farce and displacement will always be 90°. So work done equal to 0 .

Q7

$$\vec{\tau}=\vec{p}\times\vec{\mathrm{E}}$$

$$\vec{\mathrm{p}}=\mathrm{q}\vec{\ell}$$

$$\vec{\mathrm{E}}=0.2\frac{\mathrm{V}}{\mathrm{cm}}=20\frac{\mathrm{V}}{\mathrm{m}}$$

$$\vec{\mathrm{p}}=4\times(\hat{\mathrm{i}}-\hat{\mathrm{j}}+\hat{\mathrm{k}})$$

$$=(4\hat{\mathrm{i}}-4\hat{\mathrm{j}}+4\hat{\mathrm{k}})\mu\mathrm{C}-\mathrm{m}$$

$$\vec{\tau}=(4\hat{\mathrm{i}}-4\hat{\mathrm{j}}+4\hat{\mathrm{k}})\times(20\hat{\mathrm{i}})\times 10^{-6}\mathrm{Nm}$$

$$=(8\hat{\mathrm{k}}+8\hat{\mathrm{j}})\times 10^{-5}=8\sqrt{2}\times 10^{-5}$$

$$\alpha=2$$

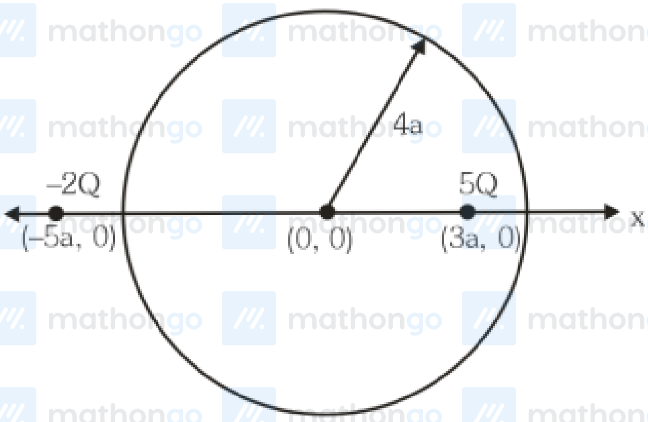
Q8

$$\text{Potential}=\frac{kQ}{R}=\frac{k.\,Ze}{R}$$

$$=\frac{9\times 10^9\times 50\times 1.6\times 10^{-19}}{9\times 10^{-13}\times 10^{-2}}$$

$$=8\times 10^6\ \mathrm{V}$$

Q9



5Q charge is inside the spherical region flux through sphere $=\frac{5Q}{\varepsilon_0}$

Q10

Ampere-Maxwell law

$\rightarrow \oint \overrightarrow{\mathrm{B}} \cdot \overrightarrow{\mathrm{dl}}=\mu_0 \mathrm{i}_{\mathrm{c}}+\mu_0 \varepsilon_0 \frac{\mathrm{d} \phi_{\mathrm{E}}}{\mathrm{dt}}$

Faraday law $\rightarrow \oint \overrightarrow{\mathrm{E}} \cdot \overrightarrow{\mathrm{d}}=\frac{\mathrm{d} \phi_{\mathrm{B}}}{\mathrm{dt}}$

Gauss' law for electricity $\rightarrow \oint \overrightarrow{\mathrm{E}} \cdot \overrightarrow{\mathrm{dA}}=\frac{\mathrm{Q}}{\varepsilon_0}$

Gauss ' law for magnetism $\rightarrow \oint \vec{B} \cdot \overrightarrow{d A}=0$

Q11

$\mathrm{u}=1 \mathrm{~m} / \mathrm{s} ; \mathrm{a}=-\frac{\sigma \mathrm{e}}{2 \varepsilon_0 \mathrm{~m}}$

$\mathrm{t}=1 \mathrm{~s}$

$\mathrm{S}=-1 \mathrm{~m}$

Using $\mathrm{S}=\mathrm{ut}+\frac{1}{2} \mathrm{at}^2$

$-1=1 \times 1-\frac{1}{2} \times \frac{\sigma \mathrm{e}}{2 \varepsilon_0 \mathrm{~m}} \times(1)^2$

$\therefore \sigma=8 \frac{\varepsilon_0 \mathrm{~m}}{\mathrm{e}}$

$\therefore \alpha=8$

Q12

$\overrightarrow{\mathrm{E}}=6\hat{\mathrm{i}}+5\hat{\mathrm{j}}+3\hat{\mathrm{k}}$
$\overrightarrow{\mathrm{A}}=30\hat{\mathrm{i}}$
$\phi=\overrightarrow{\mathrm{E}}\cdot\overrightarrow{\mathrm{A}}$
$\phi=(6\hat{\mathrm{i}}+5\hat{\mathrm{j}}+3\hat{\mathrm{k}})\cdot(30\hat{\mathrm{i}})$
$\phi=6\times30=180$

Q13

$V=\frac{kP\cos\theta}{r^2}$

& can also checked dimensionally

Q14

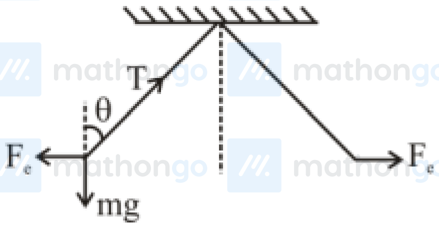
$\frac{2\mathrm{k}\lambda\mathrm{q}}{\mathrm{r}}=\mathrm{m}\omega^2\mathrm{r}$

$\omega^2=\frac{2\mathrm{k}\lambda\mathrm{q}}{\mathrm{mr}^2}$

$\left(\frac{2\pi}{\mathrm{T}}\right)^2=\frac{2\mathrm{k}\lambda\mathrm{q}}{\mathrm{mr}^2}$

$\mathrm{T}=2\pi\mathrm{r}\sqrt{\frac{\mathrm{m}}{2\mathrm{k}\lambda\mathrm{q}}}$

Q15



$\mathrm{T}\cos\theta=\mathrm{mg}$
$\mathrm{T}\sin\theta=\mathrm{F_e}$

$\tan\theta=\frac{\mathrm{F_e}}{\mathrm{mg}}$

$\tan\theta=\frac{F_e}{\rho_{\mathrm{B}}Vg}$ ...(i)

$\tan\theta=\frac{F_e}{\frac{k}{(\rho_B-\rho_L)Vg}}\ldots$ (ii)

From Eq. (i) \& (ii)

$\rho_B Vg = (\rho_B - \rho_L) kVg$
$1.4 = 0.7k$
$k = 2$

Q16

$$\left(\vec{E}_{net}\right)_P = 0$$
$$\frac{kq}{x^2} = \frac{k \cdot 3q}{(r-x)^2}$$
$$(r-x)^2 = 3x^2$$
$$r - x = \sqrt{3}x$$
$$x = \frac{r}{\sqrt{3}+1}$$

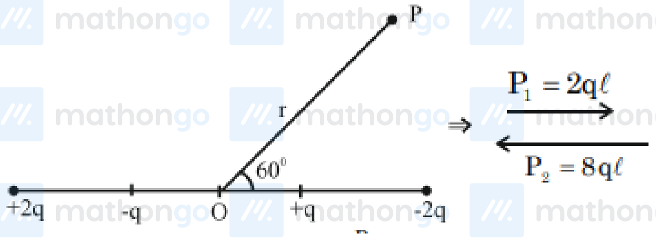
Q17

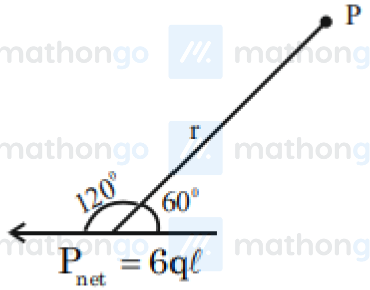
In air $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_2}$

In medium

$$F' = \frac{1}{4\pi(K\epsilon_0)} \frac{q_1 q_2}{(r')^2} = \frac{25}{4\pi(5\epsilon_0)} \frac{q_1 q_2}{(r)^2} = 5F$$

Q18





$$V = \frac{K\vec{p} \cdot \vec{r}}{r^3} = \frac{9 \times 10^9 (6q\ell)}{r^2} \cos(120^\circ)$$

$$= -(27)\left(\frac{q\ell}{r^2}\right) \times 10^9 \mathrm{Nm^2c^{-2}}$$

$$\Rightarrow \alpha = 27$$